

built once from the fault-free period (Events: also daily)

run on every new day

1-2 SENSE the building's own data, turned into events



Trend data

1

What the building-automation system already records: temperatures, flows, damper and valve positions, fan speeds, once a minute.



Configuration file

1

Maps sensor names to roles and lists the control sequence as rules. The only building-specific part: a new building means a new file, not code.



Events

2

Each day is one case. State events: what the equipment is doing. Signature events: what should not happen. Kept apart. Runs again on every new day.

3-4a UNDERSTAND what a normal day looks like, learned from one fault-free period



Process models

3

Discovered from the state events only, so they cannot learn the configured fault rules. One per terminal unit; one for the whole unit.



Calibration

4a

Healthy bands and model thresholds for every check come from the fault-free days. No threshold ever sees fault data.



Held-out days

4a

The last eight days of each month (96 a year) are kept aside and used only to measure each check's false-alarm rate.

4b CHECK eight checks score every new day, each against its healthy band

Rules

control-sequence rule broken

Residual

physical relationship drifted

Oscillation

actuator reversing too often

Unit conformance

day fits the unit model

Terminal-unit conformance

day fits the terminal-unit model

Absence

expected terminal unit silent

Frequency

daily count out of range

Rate

last 30 days against same month; support only

read the raw sensors

read the event log

5-6 ALERT only what beats its own false-alarm rate reaches the operator



Noise gate

5

A check may alarm only when its flags exceed what its own false-alarm rate would produce by chance. The rate is never taken below 3 days in 96.



Alarm to the operator

6

What broke. Where: unit, terminal unit, zone. The evidence: values and duration. The check's measured false-alarm rate.